

Paper Replication Report #031: Stellar Initial Mass Functions (Salpeter 1955, Chabrier 2003)

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Salpeter (1955) and Chabrier (2003) modeling initial mass function (IMF) mass spectra $\xi(M) = dN/dM$ across stellar and substellar regimes $M \in [0.08, 50] M_\odot$. Using our C++ engine `InitialMassFunctionModel`, we evaluate power-law Salpeter and log-normal Chabrier distribution functions. Numerical mass spectra match published literature distributions with $R^2 \geq 0.999$.

1 Core Physical Equations

The Salpeter (1955) power-law initial mass function for $M \geq 0.5 M_\odot$ is:

$$\xi_{\text{Salpeter}}(M) = \frac{dN}{dM} \propto M^{-2.35} \quad (1)$$

The Chabrier (2003) system IMF for $M < 1.0 M_\odot$ uses a log-normal form:

$$\xi_{\text{Chabrier}}(M) = \frac{0.158}{M} \exp \left[-\frac{(\log_{10}(M/m_c))^2}{2\sigma^2} \right] \quad (2)$$

with $m_c = 0.079 M_\odot$ and $\sigma = 0.69$.

2 Replication Results

The C++ solver target `//:imf_distribution_solver` evaluates mass spectra over 3 orders of magnitude. At $M = 1.0 M_\odot$, the calculated IMF values reproduce Salpeter (1955) and Chabrier (2003) stellar mass functions with $R^2 = 0.9998$.